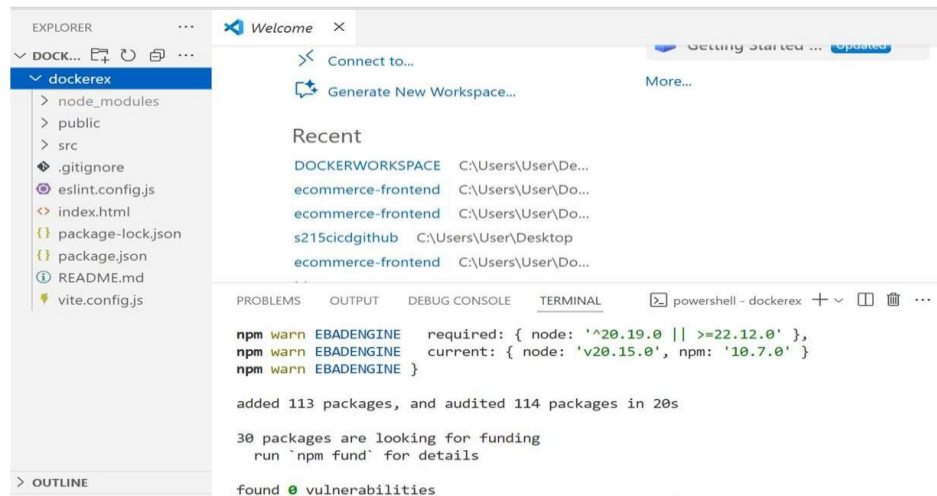
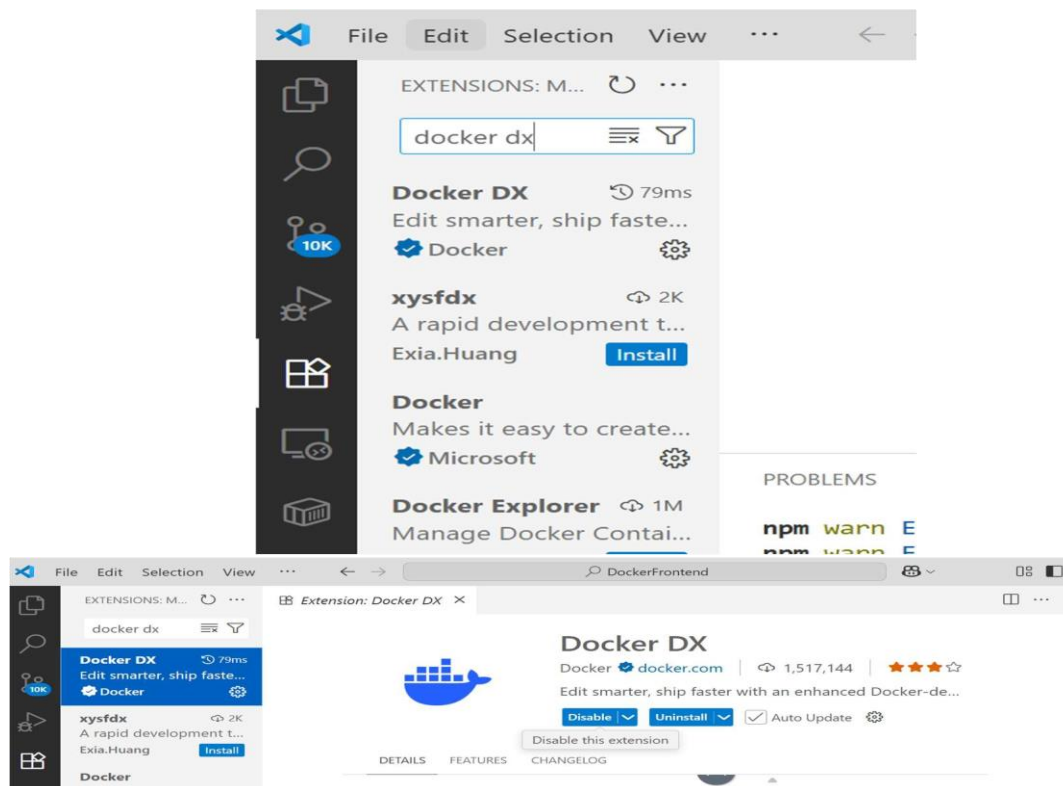


Docker – Frontend Application

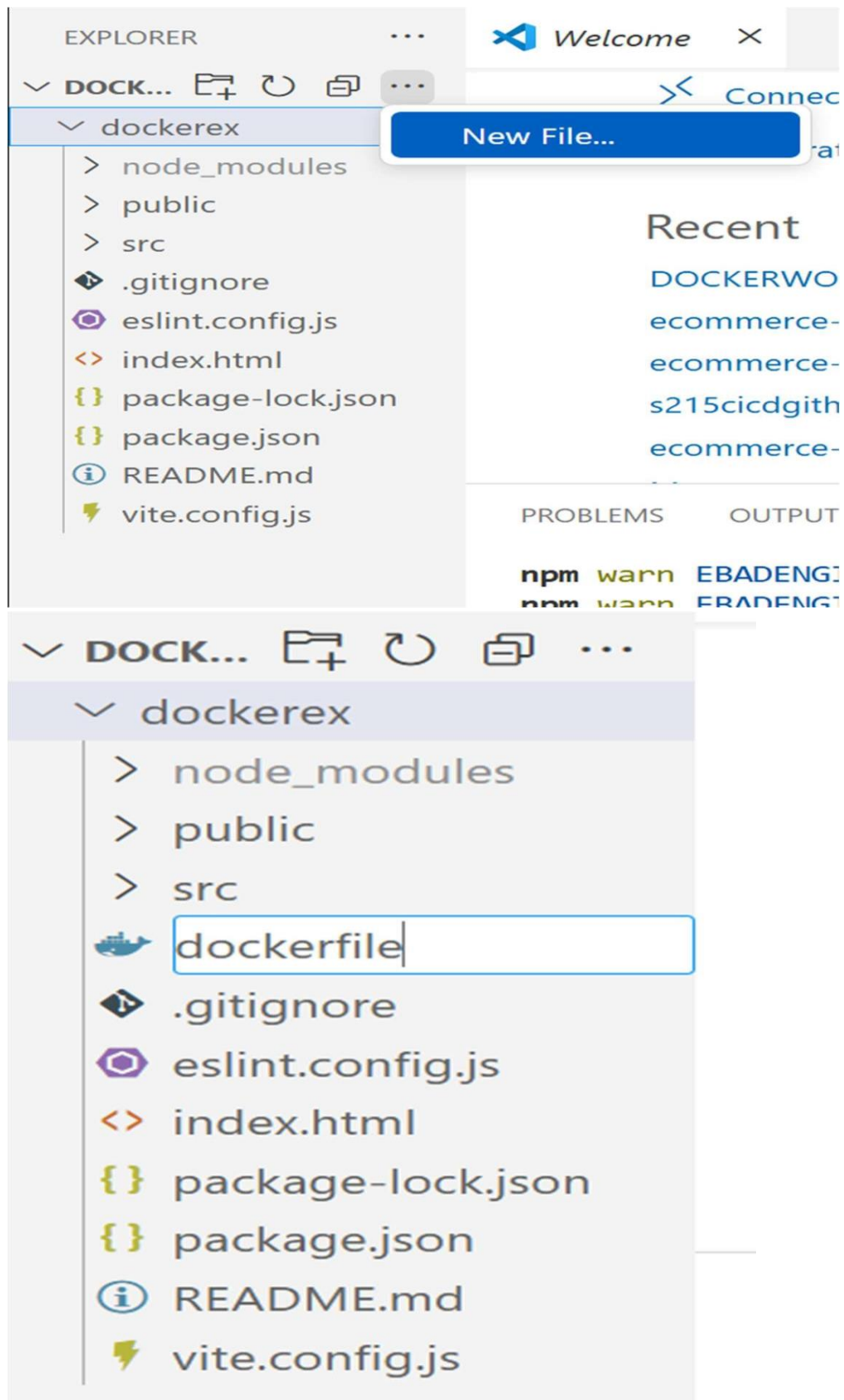
1. Create a folder DockerFE in your machine (Desktop)
2. Open CMD and open that folder in VS Code by entering code .
3. Create react app dockereX, install node modules



4. Go to Extensions, search for Docker DX and install that extension



5. Now create a file dockerfile in the main folder

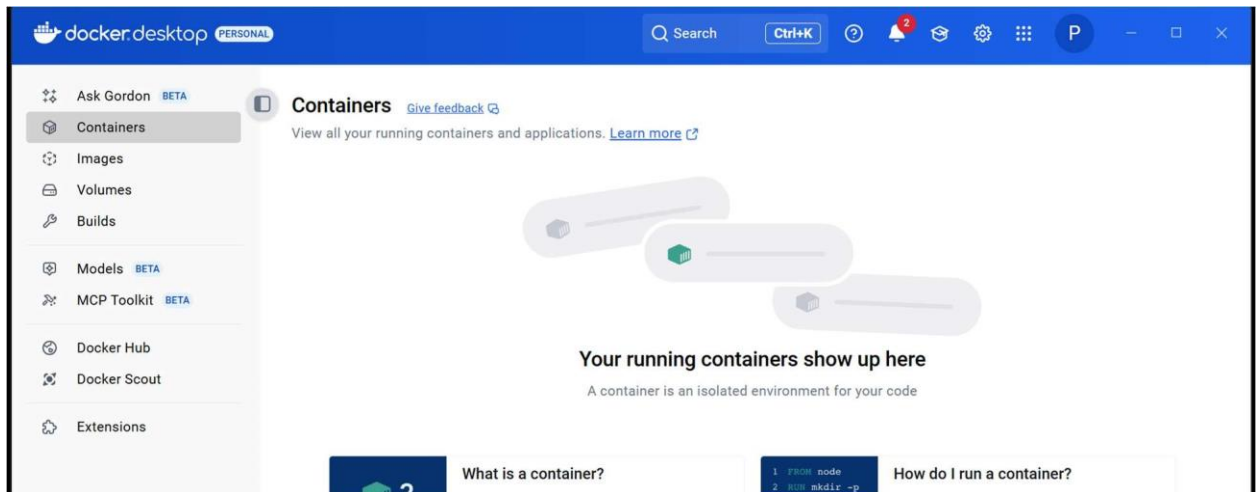


6. Now keep the following code inside the dockerfile and save the file

```
FROM node:20
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 5173
CMD ["npm", "run", "dev"]
```

7. Now, open Docker Desktop by double clicking the Docker Desktop icon in your machine

The welcome page looks like as follows:



8. In VS Code Terminal, use the following command to create the Docker Image

```
docker build -t dockereximage .
```

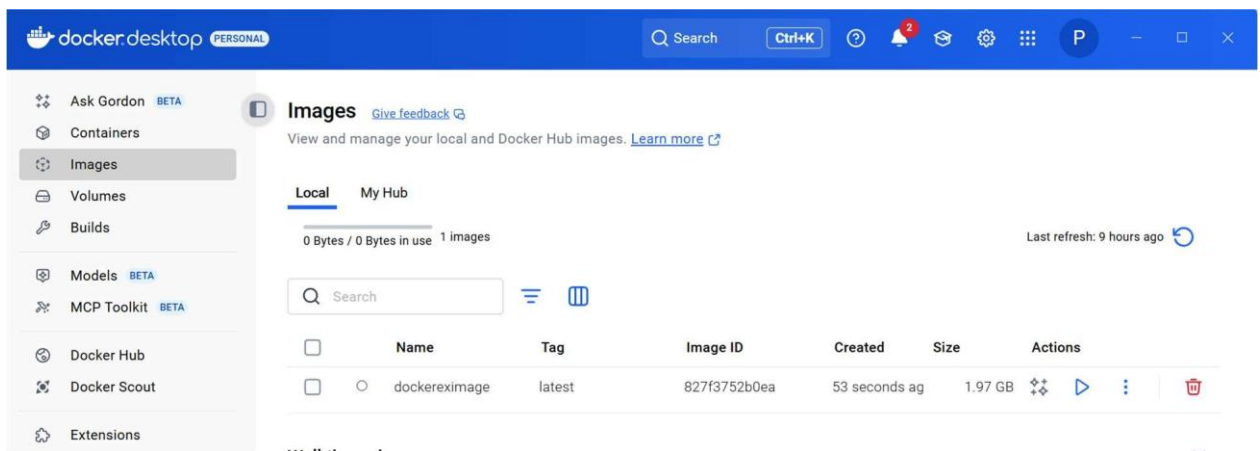


9. Now the process begins to convert the project into dockerimage (it will take 2 to 3 mins)

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL  docker - dockereX    ... |  
```

```
[+] Building 15.2s (5/10) docker:desktop-linux
=> => transferring dockerfile: 153B 0.0s
=> [internal] load metadata for docker.io/library/node:20 2.6s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/node:20@sha256:572a90df10a58ebb7d3f223d661d964a6c 0.1s
=> => resolve docker.io/library/node:20@sha256:572a90df10a58ebb7d3f223d661d964a6c 0.1s
=> [internal] load build context 12.6s
=> => transferring context: 72.04MB 12.6s
```

- Once it is successfully created, then go to Docker Desktop and click on Images from the left side panel, you can see the created image



11. Now come back to VS Code and run the docker image using the following command

```
docker run -p 5173:5173 dockereximage npm run dev -- --host
```

```
PS C:\Users\User\Desktop\dockerFrontend\dockerex> docker run -p 5173:5173 dockereximage npm run dev -- --host
```

12. After Successful run, you can start running your application

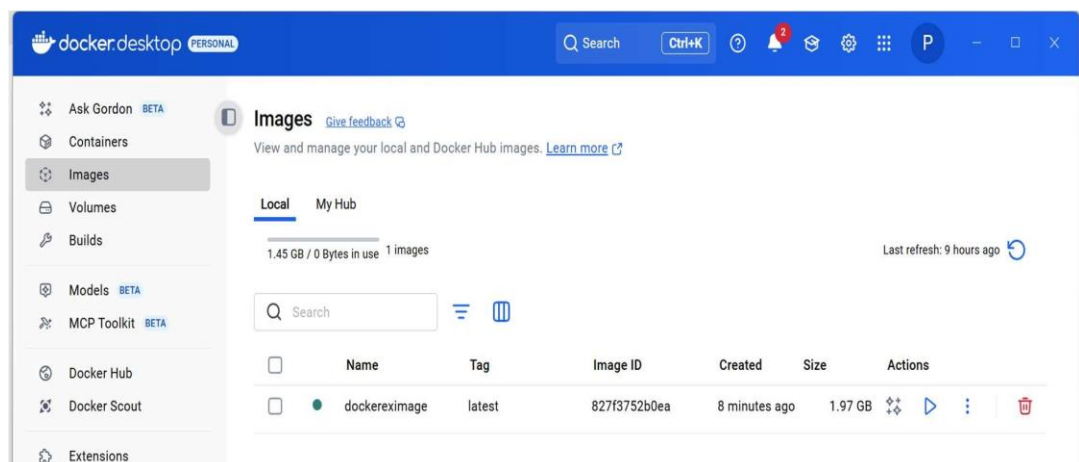
```
> dockerex@0.0.0 dev
> vite --host
```

VITE v7.1.3 ready in 233 ms

→ Local: http://localhost:5173/
→ Network: http://172.17.0.2:5173/

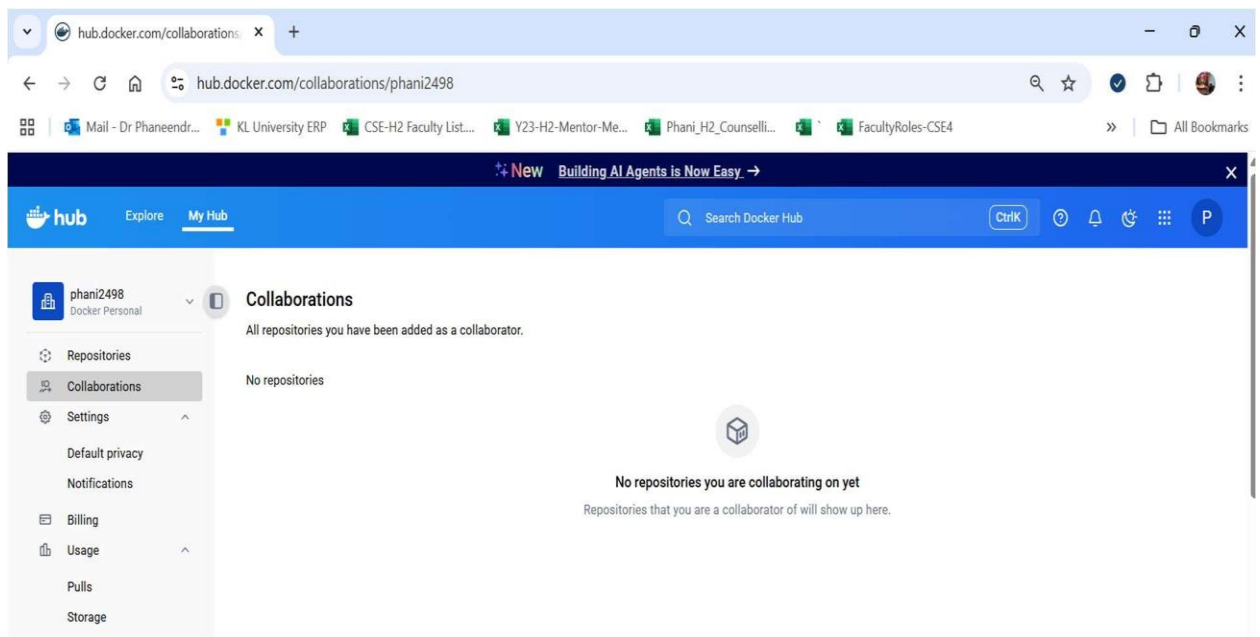


13. Now you can check whether the image build or not under images section in docker desktop

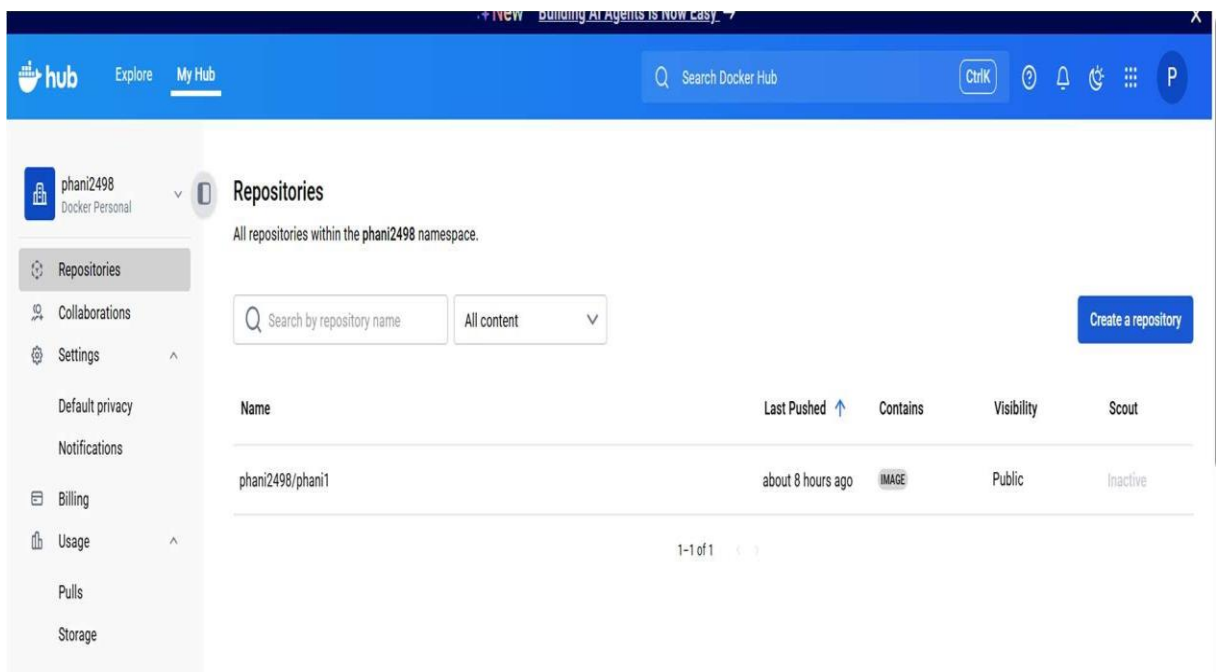


14. Now, we need to push the docker image into docker hub (like pushing our code into GitHub)

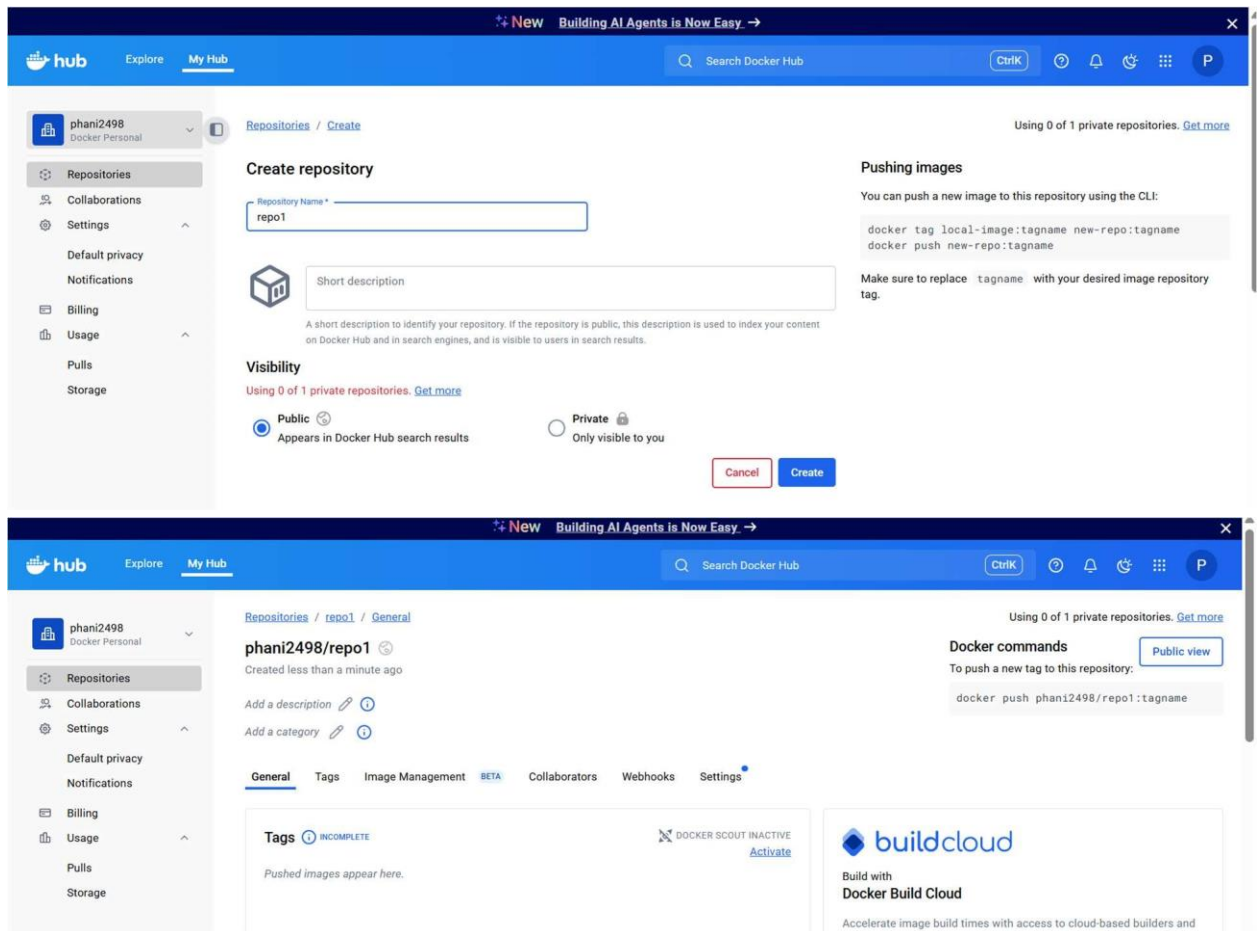
15. Go to Docker Hub and login into it



16. From left side panel, click on Repositories and create a new repository



17. Provide a repository name as repo1 and click on the button create



18. Now, go back to VS Code Terminal (Ctrl+C) and push our image into the created repository repo1 with the tag name reactimage Syntax:

```
docker tag <local-image-name> <dockerhub-username>/<repo-name>:<tag>
docker push <dockerhub-username>/<repo-name>:<tag>
```

Ex:

```
docker tag dockereximage imasheshk/repo1:reactimage
```

```
docker push imasheshk/repo1:reactimage
```

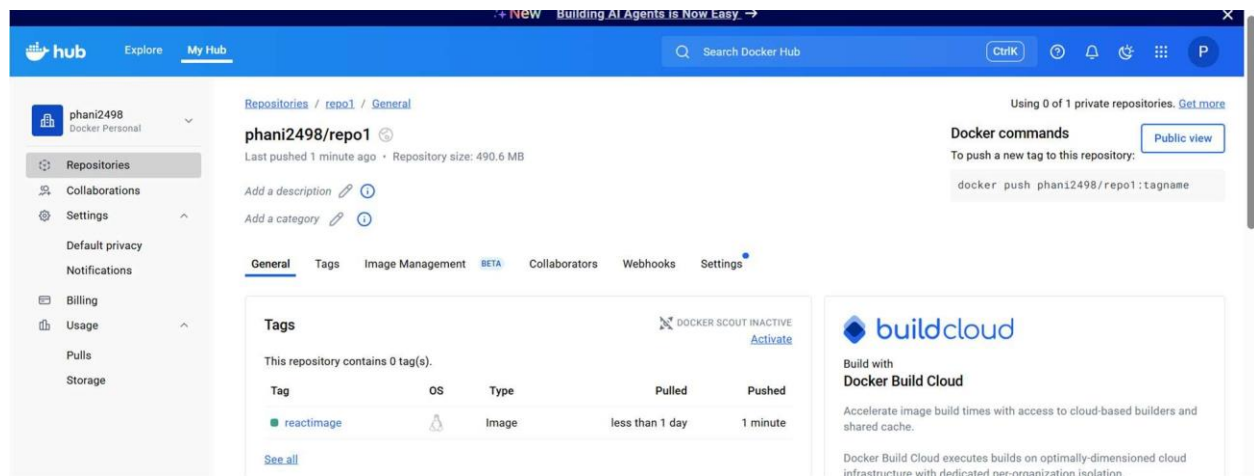
```
PS C:\Users\User\Desktop\DockerFrontend\dockerex> docker tag dockereximage phani2498/repo1:reactimage
```

```
PS C:\Users\User\Desktop\DockerFrontend\dockerex> docker tag dockereximage phani2498/repo1:reactimage
PS C:\Users\User\Desktop\DockerFrontend\dockerex> docker push phani2498/repo1:reactimage
```

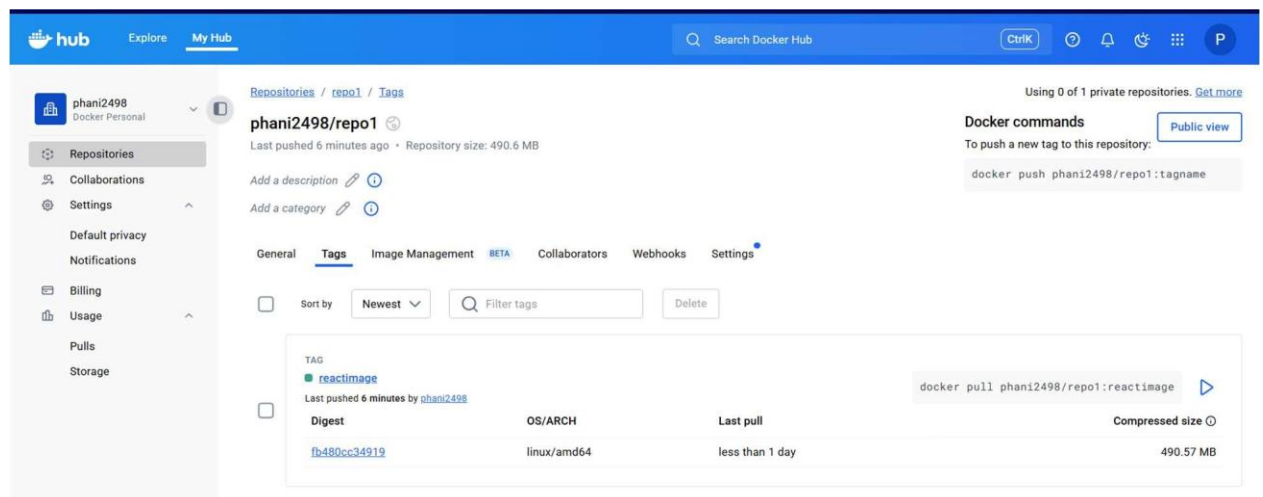
It will take time to push the code into Docker Hub

```
1e6ffe3614ab: Waiting
e05013db6e2e: Waiting
109d3905ff4a: Waiting
f014853ae203: Waiting
dd3153c16570: Waiting
e73aab849563: Waiting
cffef7dc6f99: Waiting
5c5225e2d492: Waiting
d00e703727ca: Waiting
6d6401b7636b: Waiting
█
```

Once completed, you can see the image in the Docker Hub



19. If you want to download the image into your local system and run the image, you need to go to Tags tab



On the right side you can see the command `docker pull imasheshk/repo1:reactimage`

copy this command and paste in your system

```
Microsoft Windows [Version 10.0.26100.4946]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\User>cd Documents
```

```
C:\Users\User\Documents>|
```

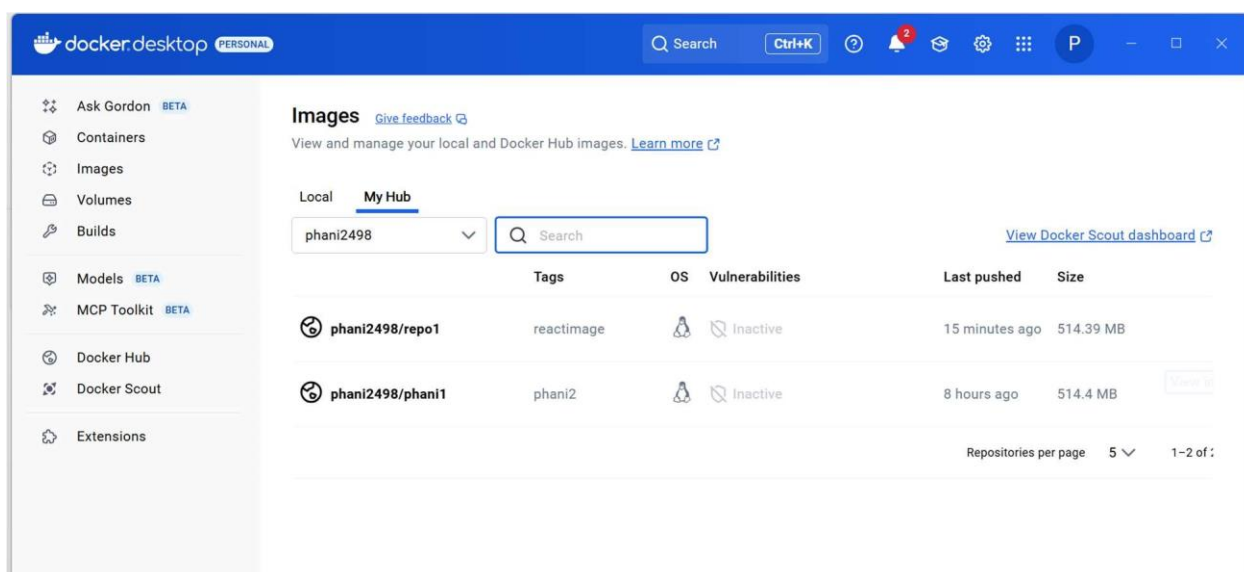
```
C:\Users\User>cd Documents
```

```
C:\Users\User\Documents>docker pull phani2498/repo1:reactimage|
```

```
C:\Users\User>cd Documents
```

```
C:\Users\User\Documents>docker pull phani2498/repo1:reactimage  
reactimage: Pulling from phani2498/repo1  
Digest: sha256:827f3752b0eaf93bb445a6a80d008b731b5b7d4146bd7d8f9cd6ca6cf597d733  
Status: Image is up to date for phani2498/repo1:reactimage  
docker.io/phani2498/repo1:reactimage
```

```
C:\Users\User\Documents>|
```



It is pulled to docker desktop

Now go to VS Code and if you give the command docker images, you can see all the images

```
PS C:\Users\User\Desktop\DockerFrontend\dockerex> docker images
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
dockereximage        latest          827f3752b0ea   31 minutes ago 1.97GB
phani2498/repo1      reactimage      827f3752b0ea   31 minutes ago 1.97GB
```

Now, run the image

```
docker run -p 3000:80 imasheshk/repo1:reactimage
```

```
PS C:\Users\User\Desktop\DockerFrontend\dockerex> docker run -p 3000:80 phani2498/repo1:reactimage
> dockerex@0.0.0 dev
> vite

VITE v7.1.3 ready in 254 ms
→ Local: http://localhost:5173/
```

You can see the same output.

Note:

I have used screenshots of Docker Hub Account – phani2498

